PROJECT DESIGN REPORT (PDR)

Project Number: 37
Project Title: Android App Development using Gen AI - Jenu Gempu (Agriculture/Self-Employment)

1. Problem Statement

Agriculture is one of the most important sectors in India, but many farmers still face challenges in accessing modern technology and expert guidance. Farmers often struggle with crop diseases, climate changes, irrigation problems, and lack of market information. Rural entrepreneurs also face difficulties in finding self-employment opportunities and business guidance.

Existing agricultural support systems are either difficult to use, available only in English, or require direct access to agricultural experts. Many farmers cannot afford professional consultation regularly. Due to these problems, crop losses and unemployment increase in rural areas.

To solve these issues, the proposed project “Jenu Gempu” aims to provide an AI-powered Android application that offers instant agricultural support, disease detection, weather-based crop suggestions, and self-employment guidance in simple language.

2. Vision

The vision of this project is to empower farmers and rural communities using modern Artificial Intelligence technologies. The application aims to make digital farming assistance easily available to everyone through smartphones.

The project focuses on improving farming productivity, reducing crop losses, encouraging self-employment, and increasing digital awareness among rural people. The app is designed to support regional languages so that farmers can use it comfortably without technical knowledge.

3. Objectives

• Develop an AI-based Android Application:
Create a user-friendly mobile application that provides smart agricultural support using Generative AI.

• Provide Smart Farming Guidance:
The app will answer farming-related questions instantly using an AI chatbot.

• Detect Crop Diseases:
Farmers can upload crop images, and AI will identify diseases and suggest solutions.

• Suggest Crops Based on Weather:
The application will analyze weather conditions and recommend suitable crops and irrigation methods.

• **Support Self-Employment:**
The app will provide ideas and guidance for rural businesses like dairy farming, poultry farming, mushroom farming, and organic farming.

• **Provide Multilingual Support:**
Users can interact with the app in Kannada and English for easy understanding.

4. Key Features & User Flow

AI Chatbot Assistance:
Users can ask agriculture-related questions through text or voice. The Generative AI chatbot provides smart and instant responses.

Crop Disease Detection:
Farmers can upload photos of crops. AI image analysis identifies diseases and suggests treatments.

Weather-Based Suggestions:
The app provides weather forecasts and recommends suitable crops based on climate conditions.

Self-Employment Guidance:
The system suggests profitable business opportunities for rural users.

Voice Support:
Users can communicate with the app using speech-to-text and text-to-speech features.

Government Scheme Information:
The app provides information about agricultural loans, subsidies, and welfare schemes.

User Flow:
1. User registers or logs into the application.
2. User selects a service such as farming support or self-employment guidance.
3. User asks questions or uploads crop images.
4. AI processes the request and generates recommendations.
5. Results are displayed to the user.
6. User can save and access previous reports.

5. Technical Requirements

Hardware Requirements:
• Intel i3 Processor or above
• Minimum 8 GB RAM
• Android smartphone with Android 8.0+
• Internet connection

Software Requirements:
• Android Studio for application development
• Kotlin or Java programming language
• Firebase for backend and authentication
• Firestore Database for cloud data storage
• Gemini API for Generative AI responses
• TensorFlow Lite for image-based disease detection
• Weather API for live weather updates

Technical Modules:
• Frontend Module – User Interface design
• Backend Module – Database and authentication
• AI Module – Chatbot and prediction system
• API Integration – Weather and speech services

6. Success Criteria

The project will be considered successful if the following conditions are achieved:

• Users can successfully register and use the application without technical difficulties.
• The AI chatbot provides accurate and meaningful agricultural guidance.
• Crop disease detection gives reliable results.
• Weather-based crop recommendations are generated properly.
• Voice interaction features work smoothly.
• The application runs with minimal crashes and errors.
• Farmers find the application easy to understand and useful in daily farming activities.

7. Impact Goals

Improve Agricultural Productivity:
Help farmers make better farming decisions using AI-based guidance.

Reduce Crop Loss:
Early disease detection and treatment recommendations help protect crops.

Promote Rural Self-Employment:
Encourage farmers and youth to start small businesses and self-employment activities.

Increase Digital Awareness:
Introduce rural communities to modern AI technologies and digital farming methods.

Support Local Language Accessibility:
Provide easy communication through Kannada and English language support.

Empower Rural Communities:
Make useful agricultural and business information accessible to everyone through smartphones.